



COMPETENCY STANDARD

FOR

**AGRICULTURE MACHINERY OPERATION AND
MAINTENANCE**

(AGRICULTURE SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for **Agriculture Machinery Operation and Maintenance** is designed to guide the development of curricula, teaching materials, and assessment tools. It also serves as a framework for providing industry-aligned training, ensuring that individuals meeting the prescribed standards through assessments are qualified to secure relevant employment opportunities.

This document has been specifically developed for training purposes under the Skills for Industry Competitiveness and Innovation Program (SICIP), aiming to address the skills requirements of the agriculture sector. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:**Generic Competencies (30 hrs.)**

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-AGR-AMOM-01-G	Apply occupational health and safety (OHS) practice in the workplace	1. Identify OHS policies and procedures. 2. Apply personal health and safety practices. 3. Report hazards and risks 4. Respond to emergencies.	10
SICIP-AGR-AMOM-02-G	Carry out workplace interaction	1 Interpret workplace communication and etiquette. 2 Read and understand workplace documents. 3 Participate in workplace meetings and discussions. 4 Practice professional ethics at work	10
SICIP-AGR-AMOM-03-G	Operate in a team environment	1 Identify team goals and work processes. 2 Identify own role and responsibilities within team. 3 Communicate and co-operate with team members. 4 Practice problem solving within the team.	10
Total Hour			30

Sector Specific Competencies (30 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-AGR-AMOM -01-S	Apply Green Practices	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
SICIP-AGR-AMOM-02-S	Work with Hand Tools and Power Tools	1. Identify and inspect hand and power tools 2. Use hand tools properly and safely 3. Operate power tools properly and safely. 4. Clean and maintain hand and power tools	20
Total Hour			30

Occupation Specific Competencies (300 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-AGR-AMOM-01-O	Apply Fundamental Knowledge of Agriculture Machinery	1. Interpret Safety Protocols for Agricultural Machinery Operation 2. Identify Agricultural Machinery 3. Understand the Key Components of Agricultural Machinery 4. Define the engine system 5. Define power transmission system	32
SICIP-AGR-AMOM-02-O	Perform Tractor (Power Tiller)	1. Prepare Tractor (Power Tiller) 2. Adjust Tractor Settings for Specific Tasks 3. Operate the Tractor (Power Tiller) 4. Monitor Tractor Performance 5. Shut Down and Store Tractor After Use	68
SICIP-AGR-AMOM-03-O	Operate Irrigation Pump Operation	1. Prepare Irrigation Pump 2. Perform Centrifugal Pump Operation 3. Perform Submersible Pump Operation	32
SICIP-AGR-AMOM-04-O	Perform Sprayer Machine	1. Prepare the Sprayer Machine 2. Operate the Manual Sprayer 3. Operate the Power Sprayer	32
SICIP-AGR-AMOM-05-O	Operate Reaper Machine	1. Set Up the Reaper Machine for Field Conditions 2. Perform Rice/Wheat Reaper Machine Operation	32
SICIP-AGR-AMOM-06-O	Perform Power Thresher (Close Drum)	1. Prepare Power Thresher 2. Operate Power Thresher 3. Maintain the Power Thresher	32

Code	Unit of Competency	Elements of Competency	Duration (hours)
		4. Shut Down and Store the Power Thresher	
SICIP-AGR-AMOM-07-O	Troubleshoot Common Engine and Mechanical Issues	1. Diagnose Starting Problems 2. Fix Gear Alignment Issues 3. Identify and Address Overheating Issues	72
Total Hours			300 Hrs.

COMPETENCY STANDARD: AGRICULTURE MACHINERY OPERATION AND MAINTENANCE

Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICE IN THE WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-AGR-AMOM-01-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying personal health and safety practices, reporting hazards and risks and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are interpreted 1.2 Safety signs and symbols are identified and followed. 1.3 Response, evacuation procedures and other contingency measures are interpreted correctly.
2. Apply personal health and safety practices	2.1 OHS policies and procedures are interpreted in the workplace including <u>personal protective equipment (PPE)</u> . 2.2 Common health issues are recognized. 2.3 Common safety issues are identified and applied.
3. Report hazards and risks	3.1 Hazards and risks are identified. 3.2 Hazards and risks assessment and controls are interpreted 3.3 Hazards and risk assessment are reported
4. Respond to emergencies	4.1 Respond to alarms and warning devices. 4.2 <u>Emergency response plans and procedures</u> are responded to. 4.3 <u>First aid procedures</u> during emergency situations are identified.

Range of variables:

Variables	Range (may include but not limited to)
1. OHS policies	1.1 Organizational OHS policies 1.2 International OHS requirements 1.3 Fire safety rules and regulations
2. Personal protective equipment	2.1 Safety glasses 2.2 Ear plugs 2.3 Gloves 2.4 Apron 2.5 Helmet 2.6 Mask 2.7 Safety shoes
3. Emergency response plans and procedures	3.1 Firefighting procedures 3.2 Earthquake response procedures 3.3 Emergency response plans and procedures 3.4 Medical and first aid
4. First aid procedure	4.1 Washing of open wound 4.2 Washing chemically infected area 4.3 Applying bandage 4.4 Taking appropriate medicine

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace OHS policies and procedures 1.2 Work safety procedures 1.3 Emergency response procedures: 1.3.1 Firefighting 1.3.2 Earthquake response 1.3.3 Accident response 1.4 Types of hazards (biological, chemical and physical) and their effects 1.5 OHS awareness 1.6 Personal protective equipment (PPE) 1.7 Malfunctions and resolutions
2. Underpinning skill	2.1 Identifying OHS policies and procedures 2.2 Applying personal health and safety practices 2.3 Reporting hazards and risks 2.4 Responding to emergencies
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace

4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace documents, signs and symbols 4.3 Codes of conduct 4.4 Projector 4.5 Learning manual 4.6 Tools, equipment and facilities appropriate to the process or activities. 4.7 Materials are relevant to the proposed activity.
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Identified OHS policies and procedures 1.2 Applied personal health and safety practices (including PPE) 1.3 Reported hazards and risks 1.4 Responded to emergencies.
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT WORKPLACE INTERACTION	Nominal Duration: 10 hrs.	Unit Code: SICIP-AGR-AMOM-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out workplace interaction. It specifically includes the tasks of interpreting workplace communication and etiquette, reading and understanding workplace documents, participating in workplace meetings and discussions and practicing professional ethics at work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret workplace communication and etiquette	1.1 Workplace codes of conduct are interpreted as per organizational guidelines. 1.2 Appropriate lines of communication are maintained with supervisors and colleagues. 1.3 Workplace interactions are conducted in a <u>courteous manner</u> to gather and convey information. 1.4 <u>Workplace procedures and matters</u> are comprehended.
2. Read and understand workplace documents	2.1 Workplace documents are interpreted correctly. 2.2 Visual information/symbols/signage are understood correctly and followed. 2.3 Specific and relevant information are accessed from <u>appropriate sources</u> . 2.4 Appropriate medium is used to transfer information and ideas.
3. Participate in workplace meetings and discussions	3.1 Team meetings are attended on time. 3.2 Meeting procedures and etiquette are followed. 3.3 Active participation is ensured, opinions are expressed and heard. 3.4 Inputs are provided and interpreted in line with the meeting purpose.
4. Practice professional ethics at work	4.1 Responsibilities as a team member are performed. 4.2 Tasks are performed in accordance with workplace procedures. 4.3 Confidentiality is maintained. 4.4 Inappropriate and conflicting situations are avoided.

Range of variables:

Variables	Range (may include but not limited to)
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1. Courteous manner	1.1 Effective questioning 1.2 Active listening 1.3 Speaking skills 1.4 Writing skill 1.5 Email etiquette
2. Workplace procedures and matters	2.1 Notes 2.2 Arranging a meeting 2.3 Agenda 2.4 Simple reports such as progress and incident reports 2.5 Job sheets 2.6 Operational manuals 2.7 Brochures and promotional material 2.8 Visual and graphic materials 2.9 Standards 2.10 OHS information 2.11 Signs
3. Appropriate sources	3.1 Human Resources (HR) Department 3.2 Managers 3.3 Supervisors 3.4 Management Information System (MIS)

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace communication and etiquette 1.2 Workplace documents, signs and symbols 1.3 Meeting procedure and etiquette 1.4 Professional ethics
2. Underpinning skill	2.1 Demonstrating workplace communication and etiquette 2.2 Interpreting workplace instructions and symbols 2.3 Demonstrating active participation in workplace meeting 2.4 Applying professional ethics at work
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process

	or activities. 4.8 Materials are relevant to the proposed activity.
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Interpreted workplace communication and etiquette 1.2 Interpreted workplace instructions and symbols 1.3 Performed active participation in workplace meetings
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE IN A TEAM ENVIRONMENT	Nominal Duration: 10 hrs.	Unit Code: SICIP-AGR-AMOM-03-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate in a team environment. It specifically includes the tasks of identifying team goals and work processes, identifying own role and responsibilities within team, communicating and cooperating with team member, and practicing problem solving within the team.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1. Roles and objectives of the team are identified and interpreted. 1.2. Roles and responsibilities of team members and work processes are identified and interpreted.
2. Identify own role and responsibilities within team	2.1. Personal role and responsibilities are identified within the team environment. 2.2. Importance of relationships within and outside the team are interpreted .
3. Communicate and co-operate with team members	3.1. Other teammates' tasks are identified and support provided when requested. 3.2. The team is encouraged through <u>sharing information</u> or expertise, working together to solve problems, and putting team success first. 3.3. Views and opinions of other team members are interpreted and respected.
4. Practice problem solving within the team	4.1. Problems faced at the individual and team level are identified and showed insight into the root-causes of the problems. 4.2. A range of solutions and courses of action are identified together with benefits, costs, and risks associated with each. 4.3. The good ideas of others to help develop solutions are recognized and advice sought from those who have solved similar problems. 4.4. It is looked beyond the obvious and not stopped at the first answers.

Range of variables:

Variables	Range (may include but not limited to)
1. Sharing information	1.1. Agenda 1.2. Minutes 1.3. Progress and incident reports 1.4. Operational manuals 1.5. Visual and graphic materials 1.6. Emails and SMS 1.7. Policy, procedure and standards 1.8. OHS information

Curricular Content Guide

1. Underpinning knowledge	1.1. Team goals and work processes 1.2. Roles and responsibilities of the team 1.3. Finding problems and solving them 1.4. Importance of views and opinions of other team members
2. Underpinning skill	2.1. Identifying own role and responsibilities within team 2.2. Communicating and co-operating with team members 2.3. Demonstrating problem solving within the team
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Eagerness to learn 3.5. Tidiness and timeliness 3.6. Environmental concerns 3.7. Respect for rights of peers and seniors at workplace 3.8. Communication with peers and seniors at workplace
4. Resource implications	The following resources must be provided: 4.1. Workplace (simulated or actual) 4.2. Workplace documents, Projector 4.3. Learning manual 4.4. Tools, equipment and facilities appropriate to the process or activities. 4.5. Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Identified own role and responsibilities within team 1.2 Communicated and co-operated with team members 1.3 Demonstrated problem solving within the team
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)

3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
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The Sector Specific Competencies

Unit of Competency: APPLY GREEN PRACTICES	Nominal Duration: 10 hrs.	Unit Code: SICIP-AGR-AMOM-01-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply green practices. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 <u>Principles of green practices</u> are explained. 1.2 <u>Sources of environmental impacts</u> during construction activities are identified. 1.3 Work activities contributing to environmental degradation, improper disposal of materials and excessive energy use are explained. 1.4 <u>Ways of mitigating environmental impacts</u> in construction sector are explained.
2. Minimize resource use in the workplace	2.1 Water, energy, and raw material consumption are documented. 2.2 <u>Recyclable and non-recyclable items</u> are identified. 2.3 Procedures to reduce resource consumption are implemented. 2.4 Sustainable alternatives to fossil-based energy resources are explored and applied.
3. Implement waste management practices	3.1 <u>Different types of waste</u> are identified. 3.2 Hazardous waste is disposed of according to environmental regulations. 3.3 Green habits to reduce waste in personal and professional life are practiced.

Range of Variables

Variable	Range
	May include but not limited to:
1. Principles of green practices	1.1 Reducing energy consumption 1.2 6R for waste management 1.2.1 Refuse 1.2.2 Reduce 1.2.3 Reuse 1.2.4 Recycle 1.2.5 Recover 1.2.6 Repair

	1.3 use of sustainable materials with low environmental impact 1.4 Recycling materials 1.5 Sustainable transportation
2. Sources of environmental impacts	2.1 Air pollution 2.1.1 Dust generation 2.1.2 Emission from machinery and vehicles 2.2 Water pollution 2.2.1 Debris and sediments 2.2.2 Chemical's leakage 2.3 Noise pollution 2.4 Waste generation 2.4.1 waste 2.4.2 Non-recyclable and hazardous materials 2.5 Resource Depletion 2.5.1 Excessive use of raw materials 2.5.2 Non-renewable material uses like use of fossil fuel, steel cement etc.
3. Ways of mitigating environmental impacts	3.1 Utilizing energy-efficient equipment 3.2 Adopting renewable energy sources 3.3 Implementing site protection measures 3.4 Using reusable materials 3.5 Choosing sustainable materials 3.6 Using noise-reducing equipment and scheduling work appropriately 3.7 Optimizing logistics and delivery
4. Recyclable and non-recyclable items	4.1 Recyclable items 4.1.1 Metal 4.1.2 Wood 4.1.3 Brick 4.1.4 Concrete 4.2 Non-recyclable items 4.2.1 Paints and coatings 4.2.2 Contaminated materials like lead paint, asbestos 4.2.3 Treated wood
5. Different types of waste	5.1 Wood waste 5.2 Metal waste 5.3 Plastic waste 5.4 Asbestos waste 5.5 Gypsum board waste 5.6 Packaging waste 5.7 Organic waste 5.8 Chemical West 5.9 Fuel/Oil west

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in shipbuilding sector 1.2 Key terms and symbols related to environmental sustainability in construction drawings 1.3 Sources of environmental impacts in construction 1.4 Methods to minimize resource consumption (water, energy, raw materials) 1.5 Waste management practices 1.6 Sustainable alternatives to fossil-based energy resources 1.7 Personal and workplace habits for reducing environmental impact
2. Underpinning Skills	2.1 Identifying environmental impacts in construction activities 2.2 Applying methods to minimize resource use in the workplace 2.3 Implementing waste management practices 2.4 Following procedures to safely dispose of hazardous materials 2.5 Documenting water, energy, and material consumption in the workplace 2.6 Performing green practices in personal and professional activities
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of construction hand tools and power tools 4.3 Safety gear 4.4 Pens 4.5 Papers 4.6 Work books 4.7 Operation and maintenance manuals 4.8 Waste segregation bins

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Explained the principles of green practices in construction sector 1.2 Identified sources of environmental impacts 1.3 Implemented procedures to minimize resource use (water, energy, raw materials) 1.4 Disposed of hazardous waste in compliance with safety and environmental standards 1.5 Practiced green habits to reduce waste in both personal and professional life
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: WORK WITH HAND AND POWER TOOLS	Nominal Duration: 20 hrs.	Unit Code: SICIP-AGR-AMOM-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to use hand and power tools in the workplace. It specifically includes the task of identifying and inspecting hand and power tools, using hand tools properly and safely, operating power tools properly and safely and cleaning and maintaining hand and power tools.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and inspect hand and power tools	1.1 Appropriate hand and power tools are identified. 1.2 Application of hand and power tools is recognized. 1.3 Usability of hand and power tools is checked and verified.
2. Use hand tools properly and safely	2.1. Appropriate <u>hand tools</u> are selected. 2.2. Safety precautions are ensured before using hand tools. 2.3. Unsafe or faulty hand tools are identified and marked for repair. 2.4. <u>Measuring tools</u> are checked and calibrated before use. 2.5. Use hand tools properly and safely to perform work activity.
3. Operate power tools properly and safely	3.1. Appropriate <u>power tools</u> are selected. 3.2. Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements. 3.3. Safety precautions are ensured before using power tools in accordance with manufacturer's operating specification. 3.4. Proper sequence of operation applied for using power tools. 3.5. Unsafe or faulty power tools are identified and marked for repair. 3.6. Operate power tools properly and safely to perform work activity.
4. Clean and maintain hand and power tools	4.1. Dust and foreign matter is removed from hand and power tools in accordance to workplace standards. 4.2. Condition of hand and power tools is checked after use and reported. 4.3. Appropriate lubricant is applied after use and prior to storage.

	4.4. Defective hand and power tools are inspected and repaired or replaced. 4.5. Hand and power tools are stored and secured in accordance with workplace requirements.
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Range of Variables

Variable	Range May include but not limited to:
1. Hand tools	1.1 Hammer 1.2 Bench vice 1.3 Files 1.4 Wrenches 1.5 Pliers 1.6 Scriber 1.7 Scraper Screw 1.8 Surface plate 1.9 Gauge 1.10 Tap sets 1.11 Die sets 1.12 Hacksaw 1.13 Spanners 1.14 Vice grip 1.15 Wire cutters 1.16 Clamps 1.17 Jacks
2. Power tools	2.1 Drills 2.2 Rivet gun 2.3 Grinders 2.4 Pneumatic wrenches 2.5 Press machine 2.6 Cutting 2.7 Saws 2.8 Soldering iron
3. Measuring tools	3.1 Micrometer 3.2 Megger 3.3 Measuring tape 3.4 Hose level 3.5 Water level 3.6 Calipers 3.7 Steel rule 3.8 Meter rule 3.9 Spirit level 3.10 Protractor 3.11 Tri-square 3.12 Gauges

Curricular Content Guide

1. Underpinning Knowledge	1.1 Information on hand and power tools, their functions and use 1.2 Procedures for safely using hand and power tools
2. Underpinning Skills	2.1 Identifying hand, power and measuring tools 2.2 Following safety precautions when using hand, power and measuring tools 2.3 Using hand and measuring tools correctly and safely in accordance with manufacturer's operating specification 2.4 Operating power tools correctly and safely in accordance with manufacturer's operating specification 2.5 Cleaning and maintaining hand and power tools after use 2.6 Applying appropriate lubricant on hand and power tools after use and prior to storing
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and selected appropriate hand and power tools for work to be performed 1.2 identified and used measuring and testing tools appropriate to work activity 1.3 followed safety precautions when using hand and power tools 1.4 operated power tools safely and pursuant to manufacturer's operating specification
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	1.5 performed cleaning and maintenance of hand and power tools after use and prior to storing
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: APPLY FUNDAMENTAL KNOWLEDGE OF AGRICULTURE MACHINERY	Nominal Duration: 32 Hrs.	Unit Code: SICIP-AGR-AMOM-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply fundamental knowledge of agriculture machinery. It specifically includes the task of interpreting safety protocols for agricultural machinery operation, identifying agricultural machinery, understanding the key components of agricultural machinery, defining the engine system and defining power transmission system.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret Safety Protocols for Agricultural Machinery Operation	1.1 <u>Personal protective equipment (PPE)</u> is identified and described. 1.2 Machine-specific safety procedures are identified. 1.3 <u>Risk assessment</u> procedures are explained. 1.4 <u>Potential hazards</u> are identified in machinery operation. 1.5 Safety protocols are identified and interpreted.
2. Identify Agricultural Machinery	2.1 <u>Common agricultural machines</u> are identified and explained. 2.2 Operating principles of agricultural machinery are understood. 2.3 <u>Safety features</u> of agricultural machinery are recognized and explained. 2.4 <u>Modern technologies</u> integrated into agricultural machinery are identified and understood.
3. Understand the Key Components of Agricultural Machinery	3.1 <u>Key components of agricultural machinery</u> are identified. 3.2 The function and role of each component of agricultural machinery are explained. 3.3 <u>Role of hydraulic systems</u> is recognized. 3.4 <u>Control systems</u> are understood for their role in the operation of the machinery.
4. Define the engine system	4.1 <u>Type of engines</u> are identified and explained. 4.2 <u>Engine systems</u> are identified and understood. 4.3 <u>Basic components of the engine system</u> are identified and defined.
5. Define power transmission system	5.1 <u>Type of power transmissions</u> are identified and described. 5.2 <u>Components of the mechanical power transmission system</u> are identified and defined.

	5.3 Function of each component is explained.
	5.4 Operating principles of the power transmission system are explained.

Range of Variables

Variable	Range (Includes but not limited to):
1. Personal protective equipment (PPE)	1.1 Apron 1.2 Mask 1.3 Hand gloves 1.4 Safety goggles 1.5 Safety helmets 1.6 Safety footwear
2. Risk assessment	2.1 Moving parts 2.2 Noise 2.3 Vibration 2.4 Entanglement 2.5 Chemical exposure
3. Potential hazards	3.1 Mechanical Hazards 3.2 Chemical Hazards 3.3 Physical hazards
4. Common agricultural machines	4.1 Tractors 4.2 Threshers 4.3 Maize sheller 4.4 Harvesters 4.5 Plows 4.6 Seeders 4.7 Sprayers 4.8 Weeder 4.9 Chopper
5. Safety features	5.1 Guards 5.2 Emergency stops 5.3 Warning signals
6. Modern technologies	6.1 GPS 6.2 Automation 6.3 Precision farming tools
7. Key components of agricultural machinery	7.1 Engine 7.2 Transmission system 7.3 Hydraulic system 7.4 Control panel
8. Role of hydraulic systems	8.1 Controlling movement 8.2 Force 8.3 Lifts 8.4 Blades 8.5 Wheels
9. Control systems	9.1 Mechanical controls

	9.2 Electronic controls 9.3 Joysticks 9.4 Manual levers
10. Type of engines	10.1 Ignition Type: 10.1.1 Spark Ignition (SI) Engines 10.1.2 Compression Ignition (CI) Engines 10.2 Combustion Cycle: 10.2.1 Four-Stroke Cycle 10.2.2 Two-Stroke Cycle
11. Engine systems	11.1 Fuel system 11.2 Cooling system 11.3 Ignition system 11.4 Lubrication system 11.5 Transmission system
12. Basic components of the engine system	12.1 Engine block 12.2 Pistons 12.3 Crankshaft 12.4 Cylinder head 12.5 Exhaust and inlet valve 12.6 Fly wheel 12.7 Oil and fuel pump 12.8 Filter
13. Type of power transmissions	13.1 Mechanical power transmission 13.2 Hydraulic power transmission
14. Components of the mechanical power transmission system	14.1 Gearbox 14.2 Drive shafts 14.3 Belts 14.4 Pulleys 14.5 Couplings 14.6 Chain sprocket

Curricular Content Guide

1. Underpinning Knowledge	1.1. Identify personal protective equipment (PPE) and describe its importance in safety. 1.2. Identify machine-specific safety procedures and explain their role in preventing accidents. 1.3. Explain risk assessment procedures for identifying potential hazards in machinery operations. 1.4. Describe safety protocols and interpret how they ensure safe operation. 1.5. Identify common agricultural machines and explain their functions. 1.6. Understand the operating principles of agricultural machinery and their safety features. 1.7. Identify and understand modern technologies integrated into agricultural machinery.
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	1.8. Define the role of hydraulic systems and understand their function in machinery operation. 1.9. Explain the components and operating principles of engine systems and power transmissions.
2. Underpinning Skills	2.1. Demonstrating identification of machine-specific safety procedures. 2.2. Applying risk assessment procedures to identify potential hazards. 2.3. Fixing unsafe conditions in machinery operations by interpreting safety protocols. 2.4. Maintaining knowledge of common agricultural machines and their operating principles. 2.5. Servicing safety features and modern technologies integrated into agricultural machinery. 2.6. Applying understanding of key components and their functions in agricultural machinery. 2.7. Demonstrating recognition of hydraulic and control systems in machinery operation. 2.8. Maintaining awareness of engine systems and power transmission types.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities. 4.2 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted Safety Protocols for Agricultural Machinery Operation 1.2 identified Agricultural Machinery 1.3 understood the Key Components of Agricultural Machinery 1.4 defined the engine system 1.5 defined power transmission system
2. Methods of Assessment	Competency should be assessed by:

	2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM TRACTOR (POWER TILLER)	Nominal Duration: 68 Hrs.	Unit Code: SICIP-AGR-AMOM-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform tractor (power tiller). It specifically includes the task of preparing tractor (power tiller), adjusting tractor settings for specific tasks, operating the tractor (power tiller), monitoring tractor performance and shut down and storing tractor after use.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare Tractor (Power Tiller)	1.1 Safety procedures are followed according to the manufacturer's guidelines. 1.2 <u>The components of the power tiller</u> are identified and checked for readiness. 1.3 Oil, fuel and water levels are checked to ensure that the power tiller. 1.4 Clutch position is checked to ensure smooth operation and control. 1.5 Drive belt tension is checked. 1.6 Wheel pressure is checked and adjusted to the recommended levels. 1.7 <u>All safety features</u> are inspected to ensure proper function.
2. Adjust Tractor Settings for Specific Tasks	2.1 The clutch position is adjusted according to job requirements. 2.2 Wheel and tine settings are adjusted based on <u>field conditions</u> to optimize tractor performance. 2.3 <u>Machine settings</u> are adjusted as needed for specific tasks. 2.4 Power take-off (PTO) speed is configured to match the attachment requirements. 2.5 Pre-operation testing is carried out as per job requirement.
3. Operate the Tractor (Power Tiller)	3.1 Clutch position is checked in neutral position. 3.2 Engine is started following proper procedures, ensuring safety and functionality. 3.3 Gear shift is operated smoothly to control tractor speed according to the task. 3.4 Steering and braking systems are checked and operated for proper response to control the tractor. 3.5 PTO (Power Take-Off) is engaged when required to operate attachments.

	3.6 The tractor's movement is controlled.
4. Monitor Tractor Performance	4.1 Engine sound and vibration are monitored to detect any irregularities or potential mechanical issues. 4.2 The color of exhaust smoke is observed regularly to assess engine performance. 4.3 Engine temperature is checked during operation. 4.4 Fuel consumption is monitored to ensure efficient use of fuel. 4.5 Tine performance is observed to ensure proper operation. 4.6 Any abnormalities in engine performance are reported for immediate attention.
5. Shut Down and Store Tractor After Use	5.1 The till gear is disengaged and set to the neutral position to ensure the tractor is safely powered down. 5.2 Engine is idled and allowed to run at low speed before shutting down. 5.3 Fuel valve is closed to prevent fuel leakage and maintain safety. 5.4 Tractor is cleaned and lubricated after use. 5.5 Tractor is parked in a designated area.

Range of Variables

Variable	Range (Includes but not limited to):
1. The components of the power tiller	1.1 Engine 1.2 Transmission system 1.3 Wheels 1.4 Controls
2. All safety features	2.1 Emergency stop 2.2 Lights 2.3 Guards
3. Field conditions	3.1 Soil type 3.2 Field slope 3.3 Crop type
4. Machine settings	4.1 Depth control 4.2 Speed 4.3 Drive belt tension

Curricular Content Guide

1. Underpinning Knowledge	1.1. Identify and check power tiller components for readiness according to safety procedures. 1.2. Ensure oil, fuel, and water levels are correct for safe operation of the power tiller. 1.3. Clutch position and adjust for smooth operation and control. 1.4. Adjust drive belt tension and wheel pressure to
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	recommended levels. 1.5. All safety features are functional before operation. 1.6. Wheel and tine settings based on field conditions to optimize performance. 1.7. Power take-off (PTO) speed to match attachment requirements. 1.8. Understand gear shift, steering, and braking systems for proper response. 1.9. Monitor engine performance, including sound, vibration, and exhaust smoke, for any irregularities. 1.10. Shut down the tractor safely by disengaging till gear, idling the engine, and closing the fuel valve.
2. Underpinning Skills	2.1. Following safety procedures according to the manufacturer's guidelines. 2.2. Checking the components of the power tiller for readiness. 2.3. Checking oil, fuel, and water levels to ensure the power tiller is ready for use. 2.4. Checking clutch position for smooth operation and control. 2.5. Checking and adjusting drive belt tension and wheel pressure to recommended levels. 2.6. Inspecting all safety features to ensure proper function. 2.7. Adjusting clutch, wheel, and tine settings based on field conditions and job requirements. 2.8. Performing pre-operation testing and checking PTO speed to match attachment needs. 2.9. Monitoring engine performance, including sound, vibration, exhaust color, and temperature.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities. 4.2 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared tractor (power tiller) 1.2 adjusted tractor settings for specific tasks 1.3 operated the tractor (power tiller) 1.4 monitored tractor performance 1.5 shut down and store tractor after use
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE IRRIGATION PUMP OPERATION	Nominal Duration: 32 Hrs.	Unit Code: SICIP-AGR-AMOM-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate irrigation pump operation. It specifically includes the tasks of preparing irrigation pump, perform centrifugal pump operation and performing submersible pump operation.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare Irrigation Pump	1.1 Safety procedures are followed as per the manufacturer's guidelines and safety standards. 1.2 <u>Type of irrigation pump</u> is identified, and its features are understood. 1.3 Preparation of the centrifugal pump is carried out by checking components. 1.4 Preparation of the submersible pump is carried out. 1.5 Engine coupling procedure is followed. 1.6 <u>Pump components</u> are checked for cleanliness, proper installation, and functionality. 1.7 Pump testing is performed to verify its operation before starting irrigation tasks.
2. Perform Centrifugal Pump Operation	2.1 Priming is performed if required to ensure that the pump is filled with water. 2.2 Engine or motor is checked for proper functionality. 2.3 Pump is started according to operational procedures. 2.4 Performance and water discharge are checked to flow rate and pressure. 2.5 Shut down procedure is followed after operation 2.6 All systems are turned off and components are cleaned.
3. Perform Submersible Pump Operation	3.1 Power supply and electrical connections are checked to ensure proper voltage and current. 3.2 Timer settings are carried out to ensure that the safety valve operates for protection. 3.3 Pump parameters are configured to operate efficiently and within safety limits. 3.4 Submersible pump is started according to operational procedures. 3.5 Performance and water discharge are monitored to ensure that the pump operates. 3.6 Shut down procedures are followed after use. 3.7 All systems are turned off and components are cleaned.

Range of Variables

Variable	Range (Includes but not limited to):
1. Type of irrigation pump	1.1 Centrifugal Pump 1.2 Submersible Pump
2. Pump components	2.1 Filters 2.2 Valves 2.3 Impeller and shaft 2.4 Casing 2.5 Bearing 2.6 Seal 2.7 Motor/engine 2.8 Staffing box 2.9 Coupling

Curricular Content Guide

1. Underpinning Knowledge	1.1. Identify the type of irrigation pump and understand its features and functionality. 1.2. Follow manufacturer's safety procedures and guidelines for proper operation. 1.3. Prepare centrifugal and submersible pumps by inspecting components for cleanliness and installation. 1.4. Engine coupling and verify pump functionality through testing before operation. 1.5. Procedures and check performance, including water discharge, flow rate, and pressure. 1.6. Power supply and electrical connections to ensure proper voltage and current. 1.7. Pump parameters, including timer settings and safety valve operation for protection. 1.8. Follow proper shutdown procedures, turning off all systems and cleaning components after use.
2. Underpinning Skills	2.1 Following safety procedures as per the manufacturer's guidelines and safety standards. 2.2 Identifying the type of irrigation pump and understanding its features. 2.3 Preparing the centrifugal pump by checking all components for readiness. 2.4 Preparing the submersible pump by performing necessary checks. 2.5 Following engine coupling procedures to ensure proper installation. 2.6 Checking pump components for cleanliness, installation, and functionality. 2.7 Performing pump testing to verify operation before starting irrigation tasks.

	2.8 Performing priming if needed to fill the pump with water. 2.9 Checking the engine or motor for proper functionality. 2.10 Starting the pump according to operational procedures. 2.11 Configuring pump parameters to operate efficiently and within safety limits.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities. 4.2 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared Irrigation Pump 1.2 performed Centrifugal Pump Operation 1.3 performed Submersible Pump Operation
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM SPRAYER MACHINE	Nominal Duration: 32 Hrs.	Unit Code: SICIP-AGR-AMOM-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform sprayer machine. It specifically includes the tasks of preparing the sprayer machine, operating the manual sprayer and operating the power sprayer machine.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare the Sprayer Machine	1.1 <u>Sprayer machine components</u> are checked. 1.2 Pressure settings are adjusted according to the manufacturer's recommendations. 1.3 Nozzles are selected and installed, based on the type of application and desired spray pattern. 1.4 Hoses and connections are inspected for leaks or blockages. 1.5 Test spraying is conducted to spray coverage, pressure, and flow rate as needed.
2. Operate the Manual Sprayer	2.1 Manual sprayer is positioned correctly for efficient spraying, with proper alignment to the target area. 2.2 Pressure is applied manually to maintain a steady flow of solution. 2.3 Sprayer nozzle is adjusted to optimizing coverage for the specific task. 2.4 <u>Solution</u> is sprayed evenly across the target area, maintaining uniform application. 2.5 Sprayer is monitored during operation to maintain functionality. 2.6 Post-operation cleaning is performed on the sprayer to remove chemical residue.
3. Operate the Power Sprayer	3.1 Fuel fillings are checked and lubricants are applied according to the manufacturer's recommendations. 3.2 <u>Ignition systems</u> are checked for proper starting of power sprayer. 3.3 Throttle or controls are adjusted for optimal pressure and flow rate. 3.4 Nozzles are adjusted for the desired spray pattern. 3.5 Solution is applied evenly across the target area. 3.6 Power sprayer is monitored for any issues. 3.7 Post-operation cleaning is performed to maintain the sprayer's functionality.

Range of Variables

Variable	Range (Includes but not limited to):
1. Sprayer machine components	1.1 Tank 1.2 Pump 1.3 Hoses 1.4 Nozzles 1.5 Piston 1.6 Bucket 1.7 Hand liver 1.8 Pressure Regulator 1.9 Filtration System 1.10 Valves
2. Solution	2.1 Insecticides 2.2 Fungicides 2.3 Herbicides 2.4 Fertilizers
3. Ignition systems	3.1 Spark plug 3.2 High tension lead 3.3 Magneto ignition

Curricular Content Guide

1. Underpinning Knowledge	1.1. Identify and check components of the sprayer machine to ensure readiness. 1.2. Pressure settings according to manufacturer recommendations for optimal performance. 1.3. Select and install nozzles based on application type and desired spray pattern. 1.4. Inspect hoses and connections for leaks or blockages to ensure proper operation. 1.5. Spraying to verify coverage, pressure, and flow rate meet requirements. 1.6. Manual sprayer correctly for efficient coverage and apply steady pressure for uniform solution flow. 1.7. Sprayer performance during operation and adjust nozzle settings to optimize coverage. 1.8. Clean the sprayer post-operation to remove chemical residue and ensure long-term functionality. 1.9. Throttle and controls for optimal pressure and spray pattern.
2. Underpinning Skills	2.1. Checking sprayer machine components for readiness. 2.2. Adjusting pressure settings according to the manufacturer's recommendations. 2.3. Selecting and installing nozzles based on the application and desired spray pattern.

	2.4. Inspecting hoses and connections for leaks or blockages. 2.5. Conducting test spraying to verify coverage, pressure, and flow rate. 2.6. Positioning the manual sprayer correctly for efficient spraying. 2.7. Applying pressure manually to maintain a steady flow of solution. 2.8. Adjusting sprayer nozzle to optimize coverage for the specific task. 2.9. Spraying the solution evenly across the target area, ensuring uniform application. 2.10. Monitoring the sprayer during operation to ensure proper functionality.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities. 4.2 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared the sprayer machine 1.2 operated the manual sprayer 1.3 operated the power sprayer
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE REAPER MACHINE	Nominal Duration: 32 Hrs.	Unit Code: SICIP-AGR-AMOM-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate reaper machine. It specifically includes the tasks of setting up the reaper machine for field conditions, and performing rice/wheat reaper machine operation.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Set Up the Reaper Machine for Field Conditions	1.1 Safety mechanisms are verified secured for safe operation. 1.2 <u>Machine settings</u> are adjusted as per job requirements. 1.3 Calibration and testing of the machine are completed. 1.4 Fuel and oil levels are checked as necessary to prepare the machine for operation. 1.5 <u>Field conditions</u> are assessed to adjust the machine settings appropriately. 1.6 Final inspection of the machine is conducted ensuring ready for fieldwork.
2. Perform Rice/Wheat Reaper Machine Operation	2.1 Safety protocols are followed, with the operator staying clear of moving parts. 2.2 Reaper machine is positioned correctly in the field with the crop rows. 2.3 Machine controls are adjusted for optimal cutting height and speed. 2.4 Field operation is started with the reaper engaging and beginning the cutting. 2.5 Operation is monitored for smooth performance to maintain consistency and prevent blockages. 2.6 Post-operation checks are performed, with the machine inspected for any damage.

Range of Variables

Variable	Range (Includes but not limited to):
1. Machine settings	1.1 Cutter 1.2 Cutter bar speed 1.3 Stubble height 1.4 Row spacing 1.5 Greasing for movable parts 1.6 Threshing and cleaning
2. Field conditions	2.1 Terrain

	2.2 Crop type 2.3 Moisture levels
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Curricular Content Guide

1. Underpinning Knowledge	1.1. Safety mechanisms to ensure the machine is secured for safe operation. 1.2. Machine settings according to job requirements and calibrate and test the machine. 1.3. Fuel and oil levels to prepare the machine for operation. 1.4. Field conditions and adjust machine settings accordingly. 1.5. Final machine inspection to ensure readiness for fieldwork. 1.6. Safety protocols and ensure the operator stays clear of moving parts. 1.7. Position the reaper machine correctly in the field, aligned with the crop rows. 1.8. Machine controls to optimize cutting height and speed. 1.9. Operation to ensure smooth performance and prevent blockages. 1.10. Machine post-operation for any damage or maintenance needs.
2. Underpinning Skills	2.1. Verifying safety mechanisms are secured for safe operation. 2.2. Adjusting machine settings according to job requirements. 2.3. Checking fuel and oil levels to prepare the machine for operation. 2.4. Assessing field conditions to adjust machine settings appropriately. 2.5. Conducting a final inspection to ensure the machine is ready for fieldwork. 2.6. Positioning the reaper machine correctly in the field with the crop rows. 2.7. Adjusting machine controls for optimal cutting height and speed. 2.8. Starting field operation, engaging the reaper to begin cutting. 2.9. Monitoring operation to ensure smooth performance and prevent blockages.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness

	3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities. 4.2 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 set up the reaper machine for field conditions 1.2 performed rice/wheat reaper machine operation
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM POWER THRESHER (CLOSE DRUM)	Nominal Duration: 32 Hrs.	Unit Code: SICIP-AGR-AMOM-06-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform power thresher (close drum). It specifically includes the tasks of preparing power thresher, operating power thresher, maintaining the power thresher and shutting down and storing the power thresher.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare Power Thresher	1.1 Safety mechanisms are verified ensuring that shields, guards in place and functional. 1.2 <u>Thresher types</u> are identified and selected based on crop type and field conditions. 1.3 <u>Power thresher components</u> are checked ensuring in good working condition. 1.4 Fuel and oil levels are checked ensuring the thresher is properly fueled and lubricated. 1.5 <u>Machine settings</u> are adjusted based on the specific crop to be harvested. 1.6 Test run is performed, ensuring that the thresher operates functioning before full operation.
2. Operate Power Thresher	2.1 Safety protocols are followed. 2.2 Power consumption and fuel levels are monitored. 2.3 Threshing performance is monitored, ensuring that the crop is separated from the straw. 2.4 Operational issues are identified and resolved. 2.5 Post-operation checks are conducted, ensuring that the thresher is clean.
3. Maintain the Power Thresher	3.1 <u>Safety components</u> are inspected in place and functional. 3.2 Engine is cleaned regularly to ensure smooth operation. 3.3 Oil, fuel levels and coolant are checked. 3.4 Air filters are cleaned or replaced ensuring proper air circulation. 3.5 Belts and pulleys are checked ensuring proper tension. 3.6 Machine is lubricated to reduce friction and wear. 3.7 General machine cleanliness is maintained.
4. Shut Down and Store the Power Thresher	4.1 Machine is turned off by following proper shutdown procedures. 4.2 Fuel and oil levels are checked to prevent damage during storage. 4.3 Air filters and engine components are cleaned to affect machine performance.

	4.4 Threshing parts are cleaned and removed.
	4.5 Machine is stored in a dry or safe location.

Range of Variables

Variable	Range (Includes but not limited to):
1. Thresher types	1.1 Axial flow 1.2 Tangential flow
2. Power thresher components	2.1 Threshing cylinder 2.2 Straw walker 2.3 Pulley 2.4 Belt 2.5 Blower 2.6 Crop inlet and outlet 2.7 Power drive gear 2.8 Side cover 2.9 Wheel
3. Machine settings	3.1 Feed rate 3.2 Speed 3.3 Threshing parameters 3.4 Temperature control with water levelling 3.5 Greasing for movable parts
4. Safety components	4.1 Protective guards and shields cover 4.2 Feeding chute with safety cover 4.3 Emergency stop mechanism 4.4 Flywheel cover 4.5 Blower and aspirator guards 4.6 Warning labels and signage 4.7 Stable frame and anti-vibration mounts

Curricular Content Guide

1. Underpinning Knowledge	1.1. Safety mechanisms ensuring shields, guards, and other safety features are in place and functional. 1.2. Identify and select thresher types based on crop and field conditions. 1.3. Power thresher components to ensure they are in good working condition. 1.4. Fuel and oil levels to ensure proper fueling and lubrication. 1.5. Machine settings according to the specific crop to be harvested. 1.6. Test run to ensure the thresher functions properly before full operation. 1.7. Threshing performance, ensuring proper separation of crop and straw.
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	1.8. Identify operational issues during threshing. 1.9. Post-operation checks and maintain cleanliness of the machine. 1.10. Engine, oil, fuel, coolant, and air filters for optimal operation. 1.11. Belts, pulleys, and other components are properly adjusted and lubricated. 1.12. Shutdown procedures and store the machine safely after use.
2. Underpinning Skills	2.1. Verifying safety mechanisms to ensure shields and guards are in place and functional. 2.2. Checking power thresher components to ensure they are in good working condition. 2.3. Checking fuel and oil levels to ensure the thresher is properly fueled and lubricated. 2.4. Adjusting machine settings based on the specific crop to be harvested. 2.5. Performing a test run to ensure the thresher functions properly before full operation. 2.6. Following safety protocols during operation. 2.7. Monitoring power consumption and fuel levels. 2.8. Monitoring threshing performance to ensure proper crop separation from straw. 2.9. Conducting post-operation checks to ensure the thresher is clean. 2.10. Inspecting safety components to ensure they are functional. 2.11. Checking belts and pulleys to ensure proper tension. 2.12. Maintaining general machine cleanliness. 2.13. Cleaning and removing threshing parts.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities. 4.2 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared power thresher 1.2 operated power thresher 1.3 maintained the power thresher 1.4 shutdown and stored the power thresher
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: TROUBLESHOOT COMMON ENGINE AND MECHANICAL ISSUES	Nominal Duration: 72 Hrs.	Unit Code: SICIP-AGR-AMOM-07-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to troubleshoot common engine and mechanical issues. It specifically includes the tasks of diagnosing starting problems, fixing gear alignment issues, and identifying and addressing overheating issues.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Diagnose Starting Problems	1.1 <u>Diagnostic tools</u> are identified and used. 1.2 <u>Starting issues</u> are identified using visual checks and diagnostic tools. 1.3 Battery voltage and connections are checked. 1.4 Fuel levels, lines, and filters are checked for blockages or defects. 1.5 Ignition system is assessed using diagnostic tools. 1.6 Control systems are checked with diagnostic tools. 1.7 Engine is tested for proper starting function post-repair.
2. Fix Gear Alignment Issues	2.1 Gear alignment is checked to verify correct positioning. 2.2 Misalignment issues are identified. 2.3 Corrective adjustments are made to restore proper gear mesh. 2.4 Lubrication is applied to ensure smooth operation. 2.5 Securing bolts and fasteners are tightened to prevent further misalignment during operation. 2.6 Gear backlash is adjusted. 2.7 Tolerance is maintained between gear teeth for efficient power transmission. 2.8 Alignment verification is performed after adjustments.
3. Identify and Address Overheating Issues	3.1 <u>Overheating issues</u> are identified using diagnostic tools. 3.2 Cooling system is inspected and checked for blockages, leaks, or damage. 3.3 Airflow is verified to maintain adequate airflow. 3.4 Engine oil is checked suitable for cooling and lubrication. 3.5 Heat exchange components are cleaned. 3.6 Post-repair testing is conducted at normal temperature levels.

Range of Variables

Variable	Range (Includes but not limited to):
1. Diagnostic tools	1.1 Multimeter 1.2 Filler gauge 1.3 Deep stick 1.4 Compression tester
2. Starting issues	2.1 Fuel tank is empty 2.2 Loose or faulty battery connection 2.3 Damaged or worn-out spark plug 2.4 Clogged air filter 2.5 Fuel contaminated with water
3. Overheating issues	3.1 Insufficient coolant level in the reservoir 3.2 Physical damage or leakage in the radiator 3.3 Worn-out or broken fan belt 3.4 Malfunctioning or damaged water pump

Curricular Content Guide

1. Underpinning Knowledge	1.1. Identify diagnostic tools to assess machine issues. 1.2. Battery voltage and connections, ensuring they are functioning properly. 1.3. Fuel levels, lines, and filters for blockages or defects. 1.4. Ignition system and control systems using diagnostic tools. 1.5. Engine post-repair to ensure proper starting function. 1.6. Gear alignment and identify any misalignment issues. 1.7. Corrective adjustments to restore proper gear mesh and apply lubrication. 1.8. Gear backlash and maintain tolerance between gear teeth for efficient transmission. 1.9. Cooling system for blockages, leaks, or damage and ensure proper airflow. 1.10. Engine oil for proper cooling and lubrication, and clean heat exchange components. 1.11. Post-repair testing to verify machine performance at normal temperatures.
2. Underpinning Skills	2.1. Checking battery voltage and connections for proper function. 2.2. Checking fuel levels, lines, and filters for blockages or defects. 2.3. Assessing the ignition system using diagnostic tools. 2.4. Checking control systems with diagnostic tools for accurate functionality. 2.5. Testing the engine for proper starting function after repairs. 2.6. Checking gear alignment to verify correct positioning. 2.7. Applying lubrication to ensure smooth operation of components.

	2.8. Tightening securing bolts and fasteners to prevent further misalignment. 2.9. Adjusting gear backlash for optimal operation. 2.10. Maintaining tolerance between gear teeth for efficient power transmission. 2.11. Performing alignment verification after adjustments. 2.12. Conducting post-repair testing at normal temperature levels.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities. 4.2 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 diagnosed starting problems 1.2 fixed gear alignment issues 1.3 identified and addressed overheating issues
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	Agriculture Machinery Operation and Maintenance
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sq) OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Computers/ Laptop	01
2.	Multimedia Projector	01
3.	Internet connectivity	01
4.	Tractor (Power Tiller)	05
5.	Tiller Attachments	05
6.	Tractor Maintenance Tools Kit	05
7.	Centrifugal Pumps (with adjustable nozzles and valves)	06
8.	Submersible Pumps (for deep well operations)	02
9.	Manual Sprayers (for small-scale operations)	10
10.	Power Sprayers (for larger-scale operations)	05
11.	Rice/Wheat Reaper Machine	02
12.	Adjustable Reaper Blades for different crop heights	02
13.	Power Threshers (Close Drum models)	02
14.	Engine Diagnostic Tools	05
15.	Diesel engine (for maintenance)	05

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Agriculture Machinery Operation and Maintenance the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.